

GRAYS REEF BUOY 41008, USA

WMO: 997010

Lat: 31.400N

Lon: 80.868W

Elev: 0

StdP: 101.33

Time Zone: -5.00 (W05)

Period: 02-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.9	4.8	-5.9	2.3	5.2	-3.6	2.8	7.7	14.7	7.7	14.1	8.3	9.7	300	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.2	30.0	27.1	29.4	26.4	28.9	26.1	27.5	28.7	27.0	28.4	26.7	28.1	4.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	22.9	28.5	26.7	22.3	28.1	26.2	21.6	27.8	87.2	28.8	84.9	28.4	83.1	28.0	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.8	11.8	10.8	DB	0.4	30.8	2.6	1.1	-1.5	31.6	-3.0	32.2	-4.5	32.8	-6.4	33.5	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.3	11.8	12.4	15.0	19.1	23.0	26.1	27.0	27.3	26.2	22.8	18.1	14.6
	DBStd	6.12	3.75	2.97	2.79	2.14	1.75	1.14	1.03	1.21	1.48	2.32	3.21	3.57
	HDD10.0	49	26	12	2	0	0	0	0	0	0	0	1	8
	HDD18.3	656	202	167	106	16	0	0	0	0	0	3	41	121
	CDD10.0	3817	82	78	158	274	403	483	528	537	485	397	243	150
	CDD18.3	1382	0	0	3	39	145	233	269	278	235	142	33	4
	CDH23.3	11106	0	0	0	4	431	2073	2818	3041	2166	551	22	0
	CDH26.7	1800	0	0	0	0	3	191	541	737	323	5	0	0
Wind	WSAvg	5.7	6.1	5.7	5.6	5.4	5.6	5.6	5.3	5.2	6.1	5.9	6.4	6.0
Precipitation	PrecAvg	1339	102	92	102	77	85	169	160	167	128	102	66	88
	PrecMax	1694	292	265	255	168	230	253	285	321	333	301	141	211
	PrecMin	1013	29	22	15	21	18	58	67	93	38	10	10	31
	PrecStd	190	55	51	57	40	51	54	47	49	62	68	39	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.0	18.5	21.6	24.0	26.9	30.1	30.6	31.0	30.1	27.3	24.8	20.8
		MCWB	18.1	17.7	19.9	22.0	23.5	26.5	28.7	27.7	26.0	24.0	23.0	19.7
	2%	DB	18.2	17.6	20.6	23.1	26.2	28.7	29.6	30.2	29.1	26.8	24.1	20.3
		MCWB	17.5	16.9	19.2	21.8	23.3	26.0	26.7	27.1	25.7	23.8	23.1	19.3
	5%	DB	17.5	16.9	19.3	22.5	25.7	28.1	29.0	29.6	28.6	26.3	23.2	19.8
		MCWB	16.8	16.1	17.8	21.2	23.1	25.5	26.2	26.7	25.5	23.6	21.7	18.9
	10%	DB	16.6	16.1	18.4	21.8	25.2	27.7	28.6	29.1	28.2	25.8	22.2	19.0
		MCWB	15.7	15.2	16.8	20.3	23.0	25.2	25.9	26.2	25.1	23.0	20.4	18.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.8	18.3	20.7	23.2	25.2	27.0	28.4	28.4	27.2	26.0	24.0	20.4
		MCDB	18.9	18.4	21.3	23.4	25.5	27.9	29.3	29.7	28.4	26.5	24.1	20.4
	2%	WB	17.9	17.4	19.9	22.5	24.6	26.5	27.4	27.8	26.7	25.1	23.3	19.8
		MCDB	18.2	17.6	20.6	22.9	25.3	27.6	28.3	29.2	28.0	26.0	23.8	19.9
	5%	WB	17.2	16.6	18.5	21.7	24.1	26.1	26.9	27.3	26.2	24.4	22.6	19.0
		MCDB	17.6	16.8	19.2	22.3	25.0	27.3	28.1	28.8	27.8	25.7	23.4	19.3
	10%	WB	16.3	15.7	17.5	20.9	23.7	25.7	26.5	26.8	25.8	23.8	20.8	18.2
		MCDB	16.8	16.0	18.1	21.5	24.8	27.0	27.9	28.4	27.5	25.3	21.6	18.6
Mean Daily Temperature Range	5% DB	MDBR	4.0	3.7	3.4	2.9	2.3	2.0	2.0	2.2	2.2	2.7	3.3	3.7
		MCDBR	3.4	3.1	3.0	2.9	2.7	2.4	2.5	2.8	2.6	2.6	2.7	3.3
		MCWBR	3.7	3.2	2.8	2.6	2.5	2.4	2.6	2.5	2.2	2.6	2.7	3.6
	5% WB	MCDBR	3.4	2.9	2.7	2.2	2.0	2.0	2.1	2.5	2.2	2.4	2.0	2.7
		MCWBR	3.7	3.1	2.7	2.2	2.1	2.5	2.5	2.6	2.1	3.1	2.5	3.1
Clear-Sky Solar Irradiance	taub	0.338	0.348	0.374	0.399	0.444	0.483	0.507	0.493	0.427	0.399	0.359	0.348	
	taud	2.532	2.509	2.452	2.372	2.288	2.235	2.174	2.217	2.394	2.442	2.505	2.527	
	Ebn at Noon	889	913	909	893	851	813	792	799	844	845	856	856	
	Edh at Noon	86	96	109	122	134	141	149	141	114	102	88	83	
All-Sky Solar Radiation	RadAvg	3.01	3.67	4.88	6.12	6.64	6.54	6.48	5.84	4.86	4.24	3.32	2.68	
	RadStd	0.20	0.41	0.46	0.39	0.37	0.51	0.34	0.38	0.44	0.42	0.21	0.20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (31 neighbors)	+0.53	+0.80	N/A	N/A	N/A	N/A	N/A	-75	+177	+128

Nomenclature: See separate page